Ram Gopal Reddy

Hyderabad, Telangana

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Training

Applied Machine Learning & Data Science | *Python,NumPy, Pandas, scikit-learn* Jun 25 – Aug 25

- Completed a 2-month internship applying real-world Python and ML workflows, covering data preprocessing, visualization, and predictive modeling to develop strong data-driven problem-solving skills.
- Built and evaluated machine learning models, achieving 80% accuracy and strong classification performance across F1-score and recall, demonstrating solid model reliability.
- Authored instructional Jupyter Notebooks with step-by-step tutorials, real-dataset examples, and visual explanations, simplifying advanced ML concepts for beginners.
- Certificate Link: **CipherSchools**

Projects

Hybrid Semantic Product Search System | *Python, FastAPI, FAISS, Vector Embeddings, NLP* Jan 26 – Feb 26

- Designed and implemented a hybrid semantic search engine combining **LLM based query** understanding with **FAISS** vector similarity search for high-performance product retrieval.
- Built a FastAPI backend that processes natural language queries, extracts structured filters (price, category, rating), and executes top-K **cosine similarity** search on precomputed embeddings.
- Implemented multi-layer caching (embedding cache + LRU query cache) reducing repeated query latency to near-millisecond response times.
- **Optimized memory** usage and search performance under low-resource environments (512MB RAM).
- Repository Link: github.com/ramgopal-reddy/Vectors_LLM_Model

Smart Parking System | *Python, React, JavaScript, Data Visualization* Aug 25 – Sep 25

- Developed a machine learning model to predict parking availability using historical occupancy data.
- Built an interactive dashboard using React and JavaScript to visualize **real-time parking data** and prediction insights.
- Integrated real-time data streams with **predictive models**, improving system responsiveness and operational reliability.
- Repository Link: github.com/ramgopal-reddy/smart-parking

Certificates / Certifications

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **Aug 2025**

Oracle — Certificate Link

Introduction to Git and GitHub **Dec 2024**

Google — Certificate Link

Skills

Languages C, C++, Java, Python, JavaScript, SQL

Technologies/Frameworks: Git, GitHub, Jupyter Notebook,REST APIs, MySQL, Vector Databases

Domain Skills: Data Structures and Algorithms, Problem-Solving, Responsive Web Design, Scripting in Python and JavaScript

Education

Lovely Professional University **2023 – Present**

Bachelor of Technology - Computer Science and Engineering — CGPA: 7.29 *Phagwara, Punjab*

Narayana College **2021 – 2023**

12th — Percentage: 94% *Hyderabad, Telangana*

City Central School **2020 – 2021**

10th — CGPA: 10.0 *Kodad, Telangana*